



ASE Partial Report

Version 1.0

SESIP-2600030-01

PSA 0632793520849

22/04/2026

NXP i.MX RT700

Version B0

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Sponsor:	NXP Semiconductors Germany GmbH.
Security target:	i.MX RT700, SESIP Security Target, Revision 0.6, NXP Semiconductors, 25 March 2026.
Certification body:	Trust CB B.V.
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Document history

Version	Date	Approved by	Changes
1.0	22/04/2026	AOM	First version of the report

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1 **References**

[GP_FST_070]	Security Evaluation Standard for IoT Platforms (SESIP) Version 1.2
[CC31p2]	Common Criteria for Information Technology Security Evaluation. Part 2: Security Functional Components Version 3.1, Revision 5
[CC31p3]	Common Criteria for Information Technology Security Evaluation. Part 3: Security Assurance Components Version 3.1, Revision 5
[CEM31]	Common Criteria for Information Technology Security Evaluation. Evaluation Methodology Version 3.1 Revision
[NXP_iMXRT700-EL_v1.0]	Evidences list, v1.0, 22 nd April 2026

2 Requirements for evidence and evaluation

- 1 The evaluation was performed on the basis of the Common Evaluation Methodology [CEM31]. The examinations conducted in this report are grouped into work units according to the [CEM31]. The following table shows the dependencies between the work units defined by the [CEM31], the Common Criteria assurance elements defined by [CC31p3] and SESIP assurance elements defined by [GP_FST_070].
- 2 An evaluator action element shall be applied to the content and presentation of evidence element. The relevant application instructions are given in the respective work units as shown below:

Evaluator Action Element	Related evaluator work units according to [CEM31]	Verdict
ASE_INT.1.1E		PASS
ASE_INT.1.1C	ASE_INT.1-1	PASS
ASE_INT.1.2C	ASE_INT.1-2	PASS
ASE_INT.1.3C	ASE_INT.1-3	PASS
	ASE_INT.1-4	PASS
ASE_INT.1.4C	ASE_INT.1-5	PASS
ASE_INT.1.5C	ASE_INT.1-6	PASS
	ASE_INT.1-7	PASS
ASE_INT.1.6C	ASE_INT.1-8	PASS
ASE_INT.1.7C	ASE_INT.1-9	PASS
ASE_INT.1.7C	ASE_INT.1-10	PASS
ASE_INT.1.2E		PASS
No element assigned	ASE_INT.1-11	PASS
ASE_OBJ.1.1E		PASS
ASE_OBJ.1.1C	-	PASS
ASE_REQ.3.1E		PASS
ASE_REQ.3.1C	-	PASS
ASE_REQ.3.2C	-	PASS
ASE_REQ.3.3C	-	PASS
ASE_REQ.3.4C	-	PASS
ASE_REQ.3.5C	-	PASS
ASE_TSS.1.1E		PASS
ASE_TSS.1.1C	ASE_TSS.1-1	PASS
ASE_TSS.1.2E		PASS
No element assigned	ASE_TSS.1-2	PASS

3 Evaluation report for ASE

3.1 List of families under the ASE class

- a) ASE_INT, ST introduction
- b) ASE_OBJ, Security objectives
- c) ASE_REQ, Security requirements
- d) ASE_TSS, TOE summary specification

3.2 Dependencies

Component	Depends on	Satisfied (YN)
ASE_TSS.1		
	ASE_REQ.1	Y (fulfilled by ASE_REQ.3)

3.3 Activity verdict

Class	Verdict
ASE	PASS

3.4 Family verdicts:

Family	Verdict
ASE_INT	PASS
ASE_OBJ	PASS
ASE_REQ	PASS
ASE_TSS	PASS

3.5 Evaluator action ASE_INT.1.1E

Verdict	PASS
---------	------

- 3 The evaluator shall confirm that the information provided meets all requirements for content and presentation of evidence.

3.5.1 Work unit ASE_INT.1-1

ASE_INT.1.1C: The ST introduction shall contain an ST reference, a TOE reference, a TOE overview and a TOE description.	
The evaluator <i>shall check</i> that the ST introduction contains an ST reference, a TOE reference, a TOE overview and a TOE description.	
Evaluator(s):	JMHO
Trainee Evaluator(s):	-
Date:	31-03-2026
Verdict:	PASS

3.5.1.1 Related Observation Reports and their status

4 None.

3.5.1.2 Work unit summary of findings

5 The ST introduction of the [ST_06] contains an ST reference, a TOE reference (Platform Reference), a TOE overview and a TOE description (Platform Overview and Description).

1.1 ST Reference

I.MX RT700, SESIP Security Target, Revision 0.6, NXP Semiconductors, 25 March 2026.

1.3 Platform Reference

I.MX RT700

Table 3. Platform Reference

Reference	Value
Platform Name and Version	I.MX RT700, B0
Platform Identification	I.MX RT735S, I.MX RT758S, I.MX RT798S
Platform Type	Microcontroller platform for IoT application
Trusted Subsystem Identification	ELE-CP (S50)

1.6 Platform Overview and Description

The i.MX RT series of crossover MCUs are part of the EdgeVerse™ [edge computing](#) platform and feature Arm® Cortex®-M cores, high performance real-time functionality and MCU usability at a cost-effective price.

The I.MX RT700 crossover MCU family includes a main-compute domain with an Arm Cortex M33 and an HiFi-4 DSP both running at 325MHz, a low-energy sense-compute domain with an Arm Cortex M33 and an HiFi-1 DSP both running at 100MHz, an NXP proprietary machine learning inference core, a high-efficiency GPU, an EdgeLock Secure Enclave (Core Profile) ELE-CP, a broad range of connectivity, 7.5 MBytes of shared low-power SRAM, and an always-on domain with real time clock and wake-up signaling on events.

The I.MX RT700 is supported by an extensive developer ecosystem, featuring [MCUXpresso software and tools](#), for an optimal microcontroller design experience.

1.6.1 Platform Security Features

The I.MX RT700 employs a security subsystem, EdgeLock Secure Enclave (Core Profile) (ELE-CP (S50), legacy names CSS or CSSv2), which together with its driver provides the following security features:

- Hardware root of trust (RoT)
- Hardware cryptography accelerators (symmetric, asymmetric, secure hash, KDF, etc.)
- True Random Number Generator (TRNG) and Deterministic Random Bit Generator (DRBG)

On top of ELE-CP (S50), I.MX RT700 provides the following security features at the SoC level:

- NXP EdgeLock™ Assurance
- ARM Trustzone enabled
- Secure boot, update and debug authentication
- Physical Unclonable Function (PUF) that can generate, store and reconstruct key sizes from 64 to 4096 bits, and directly fed to ELE-CP (S50)
- OTP-based device configuration and life cycle management
- 128 bit unique device serial number for identification (UUID)
- Secure GPIO
- Intrusion and tamper detection and response sub-system
- On-chip tamper detection for voltage level, glitch, light, temperature and reset
- Device Identifier Composition Engine (DICE)

Specific to Secure Boot and Update, I.MX RT700 supports:

- Secure boot using ECDSA P-256/384 signed images
- Custom certificate format to validated image public keys
- Up to four revocable Root of Trust (RoT) or Certificate Authority keys, Root of Trust establishment by storing the SHA-2 hash digest of the hashes of up to four RoT public keys
- Anti-rollback feature for firmware update and revocable image signing keys/certificates
- Image authentication APIs and authentication of XIP images
- Recovery booting from Serial NOR flash and ISP (UART, I2C, SPI, USB)
- SB commands to program flash, OTP-eFuse, QSPI flash programming, write to RAM and execute RAM (after image authentication)
- SB3 firmware update APIs
- Boot ROM supports Device Attestation with support of Device Identifier Composition Engine (DICE)

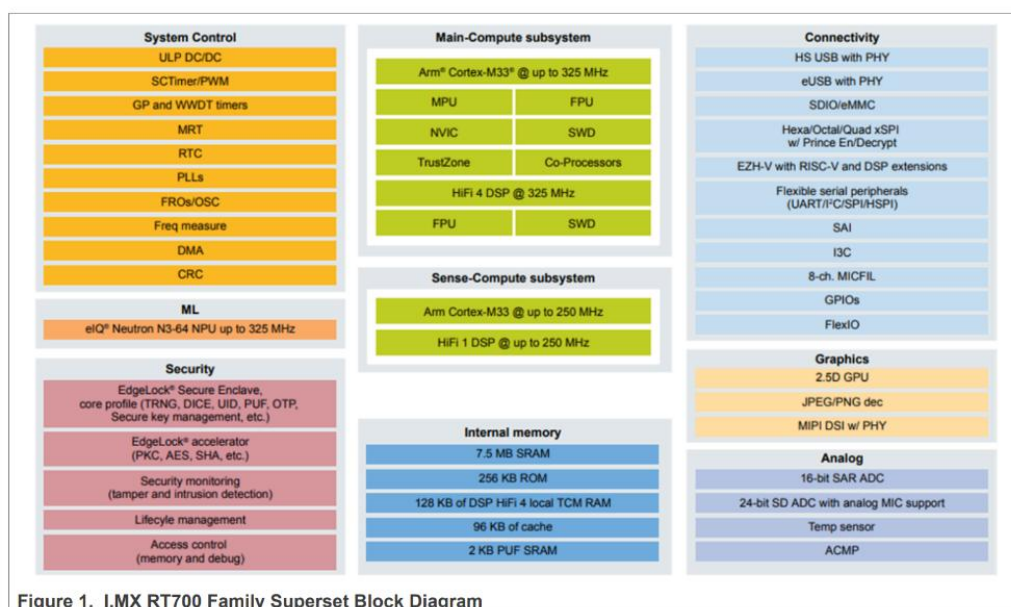
For more product feature beyond security, refer to Chapter 2 of [5].

1.6.2 Platform Physical Scope

The physical scope is the I.MX RT700 microcontroller silicon chip as shown in Figure 1.

The hardware components and interfaces are listed in Chapter 2 of [5].

The security functionalities are expressed through the SFRs listed in Section 3. .

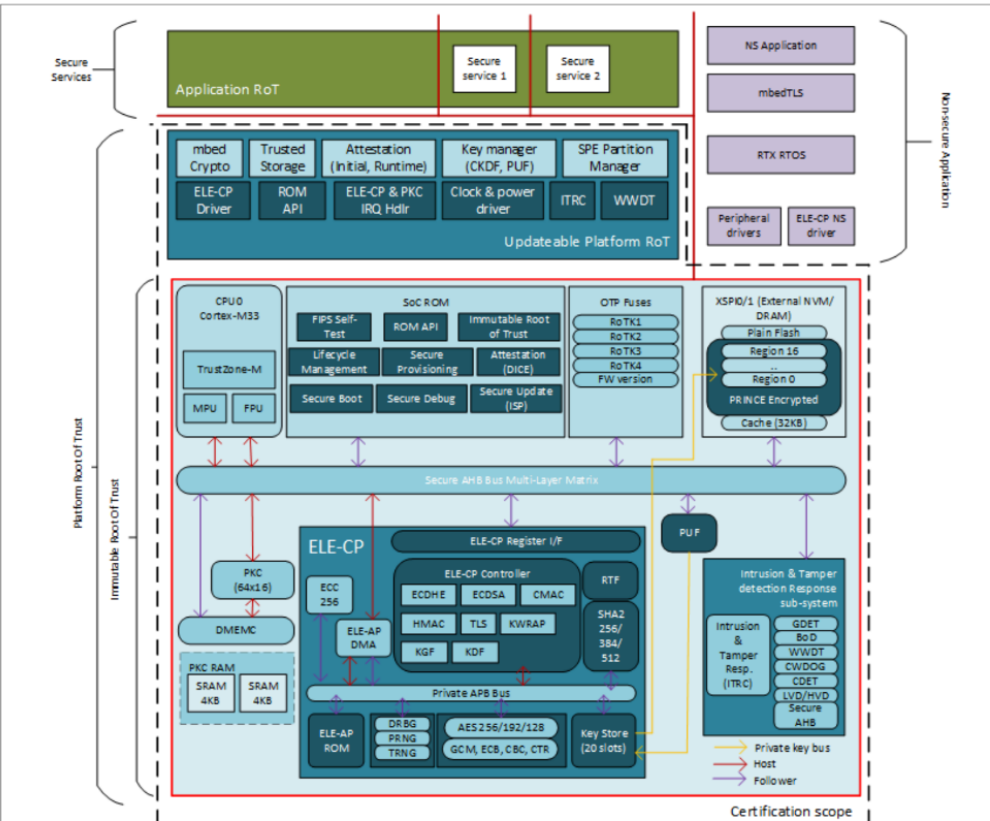


1.6.3 Platform Logical Scope

The logical scope includes the ROM firmware, and the optional flash loadable updatable platform root of trust (RoT) as illustrated in Figure 2 and listed in Table 6. Any additional firmware, OS or application software stored on the platform is not in scope of this evaluation. See Table 4 for a list of applicable guidance documents.

Table 6. Platform Deliverables			
Type	Name	Release	Form of delivery
IC Hardware	IMX RT700	B0	Silicon Chip
System ROM Firmware	IMX RT700 ROM	T2.0.32	Onchip ROM Firmware
System ROM Firmware Patch	IMX RT700 ROM Patch	N/A	Onchip Firmware
Secure Enclave	EdgeLock Secure Enclave (Core Profile) (ELE-CP (S50))	3.10.0	Onchip Hardware Subsystem
Secure Enclave ROM	ELE-CP (S50) Microcode	2.10.0	Onchip ROM Firmware
Updatable Platform RoT	ELE-CP (S50) SDK	SDK_25.12.00_MIMXRT700-EVK (RFP 2.0 B0)	Software Package
Crypto Library	Crypto Library Normal Secure for i.MX RT700 SDK	SDK_V2.0.0	included in Software Package

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14

15 The evaluator concludes that the Platform reference uniquely identifies the TOE.

3.5.1.3 Work unit verdict and rationale

16 The evaluator concludes that the ST contain an ST reference, a TOE reference, a TOE overview and a TOE description what is requested by the work unit.

17 Therefore, this work unit is fulfilled.

3.5.1.4 Vulnerability analysis notes

18 None.

3.5.2 Work unit ASE_INT.1-2

ASE_INT.1.2C: The ST reference shall uniquely identify the ST.	
The evaluator <i>shall examine</i> the ST reference to determine that it uniquely identifies the ST.	
Evaluator(s):	JMHO
Trainee Evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.5.2.1 Related Observation Reports and their status

19 None.

3.5.2.2 Work unit summary of findings

20 The ST reference of the [ST_06] uniquely identifies the ST.

1.1 ST Reference

I.MX RT700, SESIP Security Target, Revision 0.6, NXP Semiconductors, 25 March 2026.

21 The ST reference identifies each version of the ST, including a version number and date of publication.

3.5.2.3 Work unit verdict and rationale

22 The evaluator concludes that the ST reference uniquely identifies the ST.

23 Therefore, this work unit is fulfilled.

3.5.2.4 Vulnerability analysis notes

24 None.

3.5.3 Work unit ASE_INT.1-3

ASE_INT.1.3C: The TOE reference shall uniquely identify the TOE.	
The evaluator <i>shall examine</i> the TOE reference to determine that it uniquely identifies the TOE.	
Evaluator(s):	JMHO
Trainee Evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.5.3.1 Related Observation Reports and their status

25 None.

3.5.3.2 Work unit summary of findings

26 The Platform Reference and Platform Logical Scope uniquely identify the TOE.

1.3 Platform Reference

i.MX RT700

Table 3. Platform Reference

Reference	Value
Platform Name and Version	i.MX RT700, B0
Platform Identification	i.MX RT735S, i.MX RT758S, i.MX RT798S
Platform Type	Microcontroller platform for IoT application
Trusted Subsystem Identification	ELE-CP (S50)

27

1.6.3 Platform Logical Scope

The logical scope includes the ROM firmware, and the optional flash loadable updatable platform root of trust (RoT) as illustrated in [Figure 2](#) and listed in [Table 6](#). Any additional firmware, OS or application software stored on the platform is not in scope of this evaluation. See [Table 4](#) for a list of applicable guidance documents.

Table 6. Platform Deliverables

Type	Name	Release	Form of delivery
IC Hardware	i.MX RT700	B0	Silicon Chip
System ROM Firmware	i.MX RT700 ROM	T2.0.32	Onchip ROM Firmware
System ROM Firmware Patch	i.MX RT700 ROM Patch	N/A	Onchip Firmware
Secure Enclave	EdgeLock Secure Enclave (Core Profile) (ELE-CP (S50))	3.10.0	Onchip Hardware Subsystem
Secure Enclave ROM	ELE-CP (S50) Microcode	2.10.0	Onchip ROM Firmware
Updatable Platform RoT	ELE-CP (S50) SDK	SDK_25.12.00_MIMXRT700-EVK (RFP 2.0 B0)	Software Package
Crypto Library	Crypto Library Normal Secure for i.MX RT700 SDK	SDK_V2.0.0	Included in Software Package

28

- 29 The Platform Reference and Platform Logical Scope uniquely identifies the TOE, but the firmware name and version of the Secure Enclave are not specified in Table 3 under Platform Reference.

3.5.3.3 Work unit verdict and rationale

- 30 The evaluator concludes that the Platform reference uniquely identifies the TOE.
- 31 Therefore, this work unit is fulfilled.

3.5.3.4 Vulnerability analysis notes

- 32 None.

3.5.4 Work unit ASE_INT.1-4

ASE_INT.1.3C: The TOE reference shall uniquely identify the TOE.	
The evaluator <i>shall examine</i> the TOE reference to determine that it is not misleading.	
Evaluator(s):	JMHO
Trainee Evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.5.4.1 Related Observation Reports and their status

- 33 None.

3.5.4.2 Work unit summary of findings

- 34 The Platform Reference correctly identifies the TOE and is not misleading.
- 35 Also, the table 6 of the [ST_06] identifies the name of the hardware/software/firmware used by the TOE.

Table 6. Platform Deliverables

Type	Name	Release	Form of delivery
IC Hardware	i.MX RT700	B0	Silicon Chip
System ROM Firmware	i.MX RT700 ROM	T2.0.32	Onchip ROM Firmware
System ROM Firmware Patch	i.MX RT700 ROM Patch	N/A	Onchip Firmware
Secure Enclave	EdgeLock Secure Enclave (Core Profile) (ELE-CP (S50))	3.10.0	Onchip Hardware Subsystem
Secure Enclave ROM	ELE-CP (S50) Microcode	2.10.0	Onchip ROM Firmware
Updatable Platform RoT	ELE-CP (S50) SDK	SDK_25.12.00_MIMXRT700-EVK (RFP 2.0 B0)	Software Package
Crypto Library	Crypto Library Normal Secure for i.MX RT700 SDK	SDK_V2.0.0	included in Software Package

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3.5.4.3 Work unit verdict and rationale

37 The evaluator concludes that the Platform reference uniquely identifies the TOE and is not misleading.

38 Therefore, this work unit is fulfilled.

3.5.4.4 Vulnerability analysis notes

39 None.

3.5.5 Work unit ASE_INT.1-5

ASE_INT.1.4C: The TOE overview shall summarise the usage and major security features of the TOE.	
The evaluator <i>shall examine</i> the TOE overview to determine that it describes the usage and major security features of the TOE.	
Evaluator(s):	JMHO
Trainee Evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.5.5.1 Related Observation Reports and their status

40 None.

3.5.5.2 Work unit summary of findings

41 In the section 1.6.1 of the [ST_06] document describes the platform security features at SoC level.

1.6 Platform Overview and Description

The i.MX RT series of crossover MCUs are part of the EdgeVerse™ [edge computing](#) platform and feature Arm® Cortex®-M cores, high performance real-time functionality and MCU usability at a cost-effective price.

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1.6.1 Platform Security Features

The i.MX RT700 employs a security subsystem, EdgeLock Secure Enclave (Core Profile) (ELE-CP (S50), legacy names CSS or CSSv2), which together with its driver provides the following security features:

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- Hardware cryptography accelerators (symmetric, asymmetric, secure hash, KDF, etc.)
- True Random Number Generator (TRNG) and Deterministic Random Bit Generator (DRBG)

On top of ELE-CP (S50), i.MX RT700 provides the following security features at the SoC level:

- NXP EdgeLock™ Assurance

- ARM Trustzone enabled
- Secure boot, update and debug authentication
- Physical Unclonable Function (PUF) that can generate, store and reconstruct key sizes from 64 to 4096 bits, and directly fed to ELE-CP (S50)
- OTP-based device configuration and life cycle management
- 128 bit unique device serial number for identification (UUID)
- Secure GPIO
- Intrusion and tamper detection and response sub-system
- On-chip tamper detection for voltage level, glitch, light, temperature and reset
- Device Identifier Composition Engine (DICE)

Specific to Secure Boot and Update, i.MX RT700 supports:

- Secure boot using ECDSA P-256/384 signed images
- Custom certificate format to validated image public keys
- Up to four revocable Root of Trust (RoT) or Certificate Authority keys, Root of Trust establishment by storing the SHA-2 hash digest of the hashes of up to four RoT public keys
- Anti-rollback feature for firmware update and revocable image signing keys/certificates
- Image authentication APIs and authentication of XIP images
- Recovery booting from Serial NOR flash and ISP (UART, I2C, SPI, USB)
- SB commands to program flash, OTP-eFuse, QSPI flash programming, write to RAM and execute RAM (after image authentication)
- SB3 firmware update APIs
- Boot ROM supports Device Attestation with support of Device Identifier Composition Engine (DICE)

For more product feature beyond security, refer to Chapter 2 of [\[5\]](#).

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3.5.5.3 Work unit verdict and rationale

44 The evaluator concludes that the overview is clear and sufficient to give a general understanding of the usage and security features of the TOE.

45 Therefore, this work unit is fulfilled.

3.5.5.4 Vulnerability analysis notes

46 None.

3.5.6 Work unit ASE_INT.1-6

ASE_INT.1.5C: The TOE overview shall identify the TOE type.	
The evaluator <i>shall check</i> that the TOE overview identifies the TOE type.	
Evaluator(s):	JMHO
Trainee Evaluator(s):	-
Date:	31-03-2026
Verdict:	PASS

3.5.6.1 Related Observation Reports and their status

47 None.

3.5.6.2 Work unit summary of findings

48 The TOE type is described in Section 1.3 of the [ST_06].

1.3 Platform Reference

i.MX RT700

Table 3. Platform Reference

Reference	Value
Platform Name and Version	i.MX RT700, B0
Platform Identification	i.MX RT735S, i.MX RT758S, i.MX RT798S
Platform Type	Microcontroller platform for IoT application
Trusted Subsystem Identification	ELE-CP (S50)

49

3.5.6.3 Work unit verdict and rationale

50 The evaluator concludes the Platform overview does not identify the Platform Type, but it is in the Platform Reference.

51 Therefore, this work unit is fulfilled.

3.5.6.4 Vulnerability analysis notes

52 None.

3.5.7 Work unit ASE_INT.1-7

ASE_INT.1.5C: The TOE overview shall identify the TOE type.
--

The evaluator <i>shall examine</i> the TOE overview to determine that the TOE type is not misleading.	
Evaluator(s):	JMHO
Trainee Evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.5.7.1 Related Observation Reports and their status

53 None.

3.5.7.2 Work unit summary of findings

54 The TOE type described in the [ST_06] is clear and is not misleading.

3.5.7.3 Work unit verdict and rationale

55 The evaluator concludes that the TOE type is true and is not misleading.

56 Therefore, this work unit is fulfilled.

3.5.7.4 Vulnerability analysis notes

57 None.

3.5.8 Work unit ASE_INT.1-8

ASE_INT.1.6C: The TOE overview shall identify any non-TOE hardware/software/firmware required by the TOE.	
The evaluator <i>shall examine</i> the TOE overview to determine that it identifies any non-TOE hardware/software/firmware required by the TOE.	
Evaluator(s):	JMHO
Trainee Evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.5.8.1 Related Observation Reports and their status

58 None.

3.5.8.2 Work unit summary of findings

59 According to Section 1.6.4 of the [ST_06], the TOE has no internal flash.

1.6.4 Required Non-Platform Hardware/Software/Firmware

60 I.MX RT700 has no internal flash, hence compatible external non-volatile memory shall be deployed via FlexSPI for image storage with sufficient size. See Chapters 8 and 41 of [5].

3.5.8.3 Work unit verdict and rationale

61 The evaluator concludes that the TOE overview identifies any hardware/software/firmware outside the TOE required by the TOE.

62 Therefore, this work unit is fulfilled.

3.5.8.4 Vulnerability analysis notes

63 None.

3.5.9 Work unit ASE_INT.1-9

ASE_INT.1.7C: The TOE description shall describe the physical scope of the TOE.	
The evaluator <i>shall examine</i> the TOE description to determine that it describes the physical scope of the TOE.	
Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.5.9.1 Related Observation Reports and their status

64 None.

3.5.9.2 Work unit summary of finding

65 The platform physical scope can be found in Section 1.6.2 of the [ST_06].

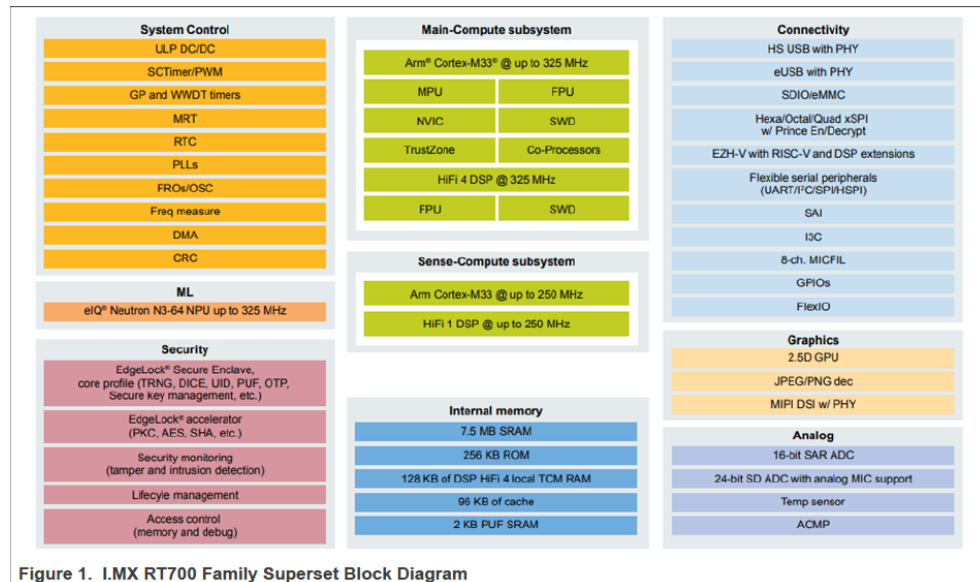
1.6.2 Platform Physical Scope

The physical scope is the I.MX RT700 microcontroller silicon chip as shown in [Figure 1](#).

The hardware components and interfaces are listed in Chapter 2 of [\[5\]](#).

The security functionalities are expressed through the SFRs listed in [Section 3](#).

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3.5.9.3 Work unit verdict and rationale

68 The evaluator concludes that the TOE description correctly describes the physical scope of the TOE.

69 Therefore, this work unit is fulfilled.

3.5.9.4 Vulnerability analysis notes

70 None.

3.5.10 Work unit ASE_INT.1-10

ASE_INT.1.8C: The TOE description shall describe the logical scope of the TOE.	
The evaluator <i>shall examine</i> the TOE description to determine that it describes the logical scope of the TOE.	
Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.5.10.1 Related Observation Reports and their status

71 None.

3.5.10.2 Work unit summary of findings

72 The platform logical scope can be found in Section 1.6.3 of the [ST_06].

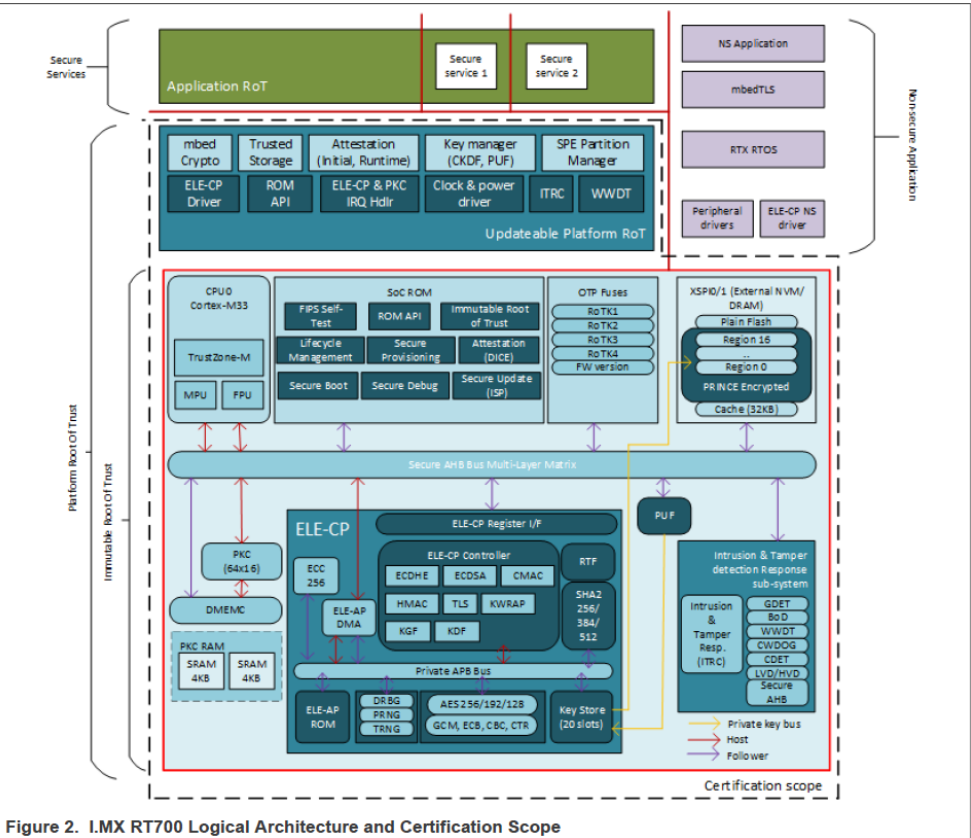
1.6.3 Platform Logical Scope

The logical scope includes the ROM firmware, and the optional flash loadable updatable platform root of trust (RoT) as illustrated in Figure 2 and listed in Table 6. Any additional firmware, OS or application software stored on the platform is not in scope of this evaluation. See Table 4 for a list of applicable guidance documents.

Table 6. Platform Deliverables

Type	Name	Release	Form of delivery
IC Hardware	i.MX RT700	B0	Silicon Chip
System ROM Firmware	i.MX RT700 ROM	T2.0.32	Onchip ROM Firmware
System ROM Firmware Patch	i.MX RT700 ROM Patch	N/A	Onchip Firmware
Secure Enclave	EdgeLock Secure Enclave (Core Profile) (ELE-CP (S50))	3.10.0	Onchip Hardware Subsystem
Secure Enclave ROM	ELE-CP (S50) Microcode	2.10.0	Onchip ROM Firmware
Updatable Platform RoT	ELE-CP (S50) SDK	SDK_25.12.00_MIMXRT700-EVK (RFP 2.0 B0)	Software Package
Crypto Library	Crypto Library Normal Secure for i.MX RT700 SDK	SDK_V2.0.0	included in Software Package

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3.5.10.3 Work unit verdict and rationale

75 The evaluator concludes that the TOE description correctly describes the logical scope of the TOE.

76 Therefore, this work unit is fulfilled.

3.5.10.4 Vulnerability analysis notes

77 None.

3.6 Evaluator action ASE_INT.1.2E

Verdict	PASS
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78 The evaluator shall confirm that the TOE reference, the TOE overview, and the TOE description are consistent with each other.

3.6.1 Work unit ASE_INT.1-11

ASE_INT.1-11: The evaluator <i>shall examine</i> the TOE reference, TOE overview and TOE description to determine that they are consistent with each other.	
Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.6.1.1 Related Observation Reports and their status

79 None.

3.6.1.2 Work unit summary of findings

80 None.

3.6.1.3 Work unit verdict and rationale

81 The evaluator after examining [ST_06] determines that the Platform Reference, Platform Overview and Platform Description are consistent with each other.

82 Therefore, this work unit is fulfilled.

3.6.1.4 Vulnerability analysis notes

83 None.

3.7 Evaluator action ASE_OBJ.1.1E

Verdict	PASS
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- 84 The evaluator shall confirm that the information provided meets all requirements for content and presentation of evidence.

3.7.1 Work unit ASE_OBJ.1-1

ASE_OBJ.1.1C: The statement of security objectives shall describe the security objectives for the operational environment.	
Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.7.1.1 Related Observation Reports and their status

- 85 None.

3.7.1.2 Work unit summary of findings

- 86 The security objectives for the operational environment are defined in Section 2.1 of the [ST_06].

2.1 Platform Objectives for the Operational Environment

For the platform to fulfill its security requirements, the operational environment (technical or procedural) must fulfill the following objectives:

Table 7. Platform Objectives for the Operational Environment

Title	Description	Reference
Follow Secure Guidance	Users shall ensure secure and correct use of the platform according to guidance listed in Section 1.4 .	This document Section 1.4
Platform Verification	The operating system or application code is expected to verify the correct version of all platform components that it depends on, as described in Section 3.2.1.1 of this document.	This document Section 3.2.1.1
Key Management	Cryptographic keys and certificates outside of the Platform are subject to secure key management procedures.	This document
OEM Trust Provisioning	Any secret to be provisioned into the platform is generated securely (e.g., via a standard compliant HSM) and subject to secure key management procedures. The provisioning process is done in secure sites with physical, logical security and organizational policies in place.	This document
Trusted Users	Actors in charge of platform management, for instance for signature of firmware update, are trusted.	This document
Secure Boot	The operating system or application code are expected to make use of the Secure Boot feature as described in Chapter 7 of [6]	Chapter 7 of [6] and [9]
Secure Update and Key Revoke	Actors in charge of executing update of the platform firmware or applications are expected to securely initiate the update process. The update image is expected to be properly signed and distributed in secure manner to ensure its confidentiality and authenticity. The operating system or application code are expected to update an image with proper remedy solution and version increased and/or revoke key in case of security incidence occurrence of the image and/or the key.	Chapter 7 of [6]
Secure Debug	The integrating environment is expected to configure the debug functionality as described in Chapter 10 of [6] and [8] to meet the extra physical attacker resistance.	Chapter 10 of [6] , and [8]
SW Integration	The operating system or application code are expected to ensure the correct version of SDK drivers are integrated and configured.	This document
Lifecycle Management	The operating system or application code are expected to provide lifecycle states and secure mechanism of lifecycle state transition according to the use case, and the operational environment is expected to configure the platform accordingly for lifecycle state transitions. In general, the operating system or application code are expected to configure the platform to In-field or In-field locked state.	Chapter 3 of [6]
Software isolation	i.MX RT700 provides two different isolation mechanisms: ELE-CP (S50) vs rest of SoC, and TrustZone Secure vs Non-secure. The operating system or application code are expected to configure and utilize at least one mechanism for isolation between platform and application.	
Physical attacker resistance configurations	If local physical attack is applicable for the use cases, the following configurations shall be kept after boot by the operating system or application code: The operating system or application code are expected to configure the <code>compute_main_clk</code> to one of the internal clock sources;	This document, Chapters 12, 13, 14 and 24 of [6]

87

Table 7. Platform Objectives for the Operational Environment...continued

Title	Description	Reference
	- The operating system or application code are expected to configure the appropriate trigger source events in the ITRC controller; - The operating system or application code are expected to enable the glitch detectors - ARM CM33 CPU is not hardened against physical attacks, e.g., voltage glitching or EMFI. It is therefore recommended to harden secure application against such attacks using software-based countermeasures and leverage code watchdog offered by i.MX RT700;	

88

3.7.1.3 Work unit verdict and rationale

89 The evaluator has checked the security objectives for the operational environmental are defined.

90 Therefore, this work unit is fulfilled.

3.7.1.4 Vulnerability analysis notes

91 None.

3.8 Evaluator action ASE_REQ.3.1E

Verdict	PASS
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92 The evaluator shall confirm that the information provided meets all requirements for content and presentation of evidence.

3.8.1 Work unit ASE_REQ.3-1

ASE_REQ.3.1C: The statement of security requirements shall describe the SFRs and the SARs.	
Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.8.1.1 Related Observation Reports and their status

93 None.

3.8.1.2 Work unit summary of findings

94 In section 3.1 of the [ST_06] is described the SARs

3.1 Security Assurance Requirements

The claimed assurance requirements packages are the following as defined in Chapter 4 of GlobalPlatform Technology Security Evaluation Standard for IoT Platforms (SESIP), version 1.2 [\[1\]](#):

- **SESIP Assurance Level 3 (SESIP3)** .

95

96 In the section 3.2 [ST_06] is described the SFRs

3.2 Security Functional Requirements

In the following Security Functional Requirements, the term **platform** covers the **i.MX RT700 physical and logical scope**, and the term **application** refers to any additional firmware, OS or application software which is out of evaluation scope. It represents a part of the final connected device.

i.MX RT700 fulfils the following security functional requirements:

97

3.8.1.3 Work unit verdict and rationale

98 The evaluator confirms that the SFRs and the SARs are described.

99 Therefore, this work unit is fulfilled.

3.8.1.4 Vulnerability analysis notes

100 None.

3.8.2 Work unit ASE_REQ.3-2

ASE_REQ.3.2C: All SFRs shall be drawn from the list of allowed Security Functional Requirements.	
Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.8.2.1 Related Observation Reports and their status

101 None.

3.8.2.2 Work unit summary of findings

102 The Security Functional Requirements collected in the [ST_06] are

SFRs for SoC	SESIP v1.2
Verification of Platform Identity	3.1.1
Verification of Platform Instance Identity	3.1.2
Attestation of Platform Genuineness	3.1.3
Secure Initialization of the Platform	3.1.4
Attestation of Platform State	3.1.5
Attestation of Application Genuineness	3.1.6
Attestation of Application State	3.1.7
Secure Update of Platform	3.2.3
Decommission of Platform	3.2.6
Field Return of Platform	3.2.7
Physical Attacker Resistance	3.4.2
Software Attacker Resistance (between SPE and NSPE)	3.4.3
Software Attacker Resistance (between PSA-RoT and Application Root of Trust Services)	3.4.3
Software Attacker Resistance: Isolation of Platform Parts	3.4.4
Cryptographic Operation	3.5.1

Cryptographic Operation (PKC)	3.5.1
Cryptographic Key Generation	3.5.2
Cryptographic KeyStore	3.5.3
Secure Data Serialization	3.6.4
Residual Information Purging	3.6.5
Residual Information Purging (SoC)	3.6.5
Reliable Index	3.6.7
Secure Debugging	3.6.8
Secure Recovery	3.6.9

103 All SFRs are defined in [GP_FST_070]

3.8.2.3 Work unit verdict and rationale

104 The evaluator concludes that all SFRs are allowed.

105 Therefore, this work unit is fulfilled.

3.8.2.4 Vulnerability analysis notes

106 None.

3.8.3 Work unit ASE_REQ.3-3

ASE_REQ.3.3C: The SFR “Verification of Platform Identity” shall be included. The SFR “Secure Update of Platform” shall be included or under the ALC_FLR.2 it shall be argued why updates are not applicable.	
Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.8.3.1 Related Observation Reports and their status

107 None.

3.8.3.2 Work unit summary of findings

108 The SFR “Verification of Platform Identity” is included in the [ST_06].

3.2.1.1 Verification of Platform Identity

The platform provides a unique identification of the platform, including all its parts and their versions.

Conformance rationale:

109

Conformance rationale:

The ELE-CP Hardware and Software (microcode) versions are available from `ELS_VERSION` register. The values shall match the versions indicated in [Section 1.6.3](#). See more in Chapters 17.3.2.22 and 17.4.21 of [\[6\]](#).

110

The production validation process ensures that the SFR behaves correctly.

The ELE-CP (S50) Hardware and Software (microcode) versions are available from `ELS_VERSION` register as per chapters 17.3.2.22 and 17.4.21 of [\[6\]](#) and shall match the versions indicated in [Section 1.6.3](#).

The ROM version can be read using the `GetProperty` command in ISP mode with tag 18h (Target ROM build version) as per chapters 9.5.2 and 9.5.16 of [\[6\]](#) and shall match the value in [Section 1.6.3](#).

The ROM patch version can be read from `ELS_AS_CFG3` register as per chapter 13.6.1.72 of [\[6\]](#) and shall match the value in [Section 1.6.3](#).

The SoC Hardware version can be read using the `GetProperty` command in ISP mode with tag 10h as per chapters 9.5.2 and 9.5.16 of [\[6\]](#) and shall match the value in [Section 1.6.3](#).

The CryptoLib version can be read using the `mcuxCl_GetVersion()` API. See "mcuxCl_GetVersion" in "mcuxCl_cls.h" section of [\[11\]](#).

111

The production validation process ensures that the SFR behaves correctly.

112

The SFR "Secure Update of Platform" is included in the [\[ST_06\]](#).

3.2.2.1 Secure Update of Platform

The platform can be updated to a newer version in the field such that the *confidentiality*, integrity and authenticity of the platform is maintained.

Conformance rationale:

The secure part of the ROM bootloader of i.MX RT700 provides a secure firmware update operation. Secure Update is the process used to securely update the firmware image in the field. The firmware image is encrypted using AES-128 or AES-256 and signed using ECDSA P-256, following the SB3.1 firmware image format. Secure Update guarantees authenticity and confidentiality of the new image. It also ensures that the new image is up-to-date, preventing the rollback to an older image. Running firmware is in charge of receiving and verifying the new firmware image. The follow-up Secure Boot verifies the new firmware image again, making sure the Immutable RoT is still in charge of ensuring authenticity of the latest firmware.

The anti-rollback is achieved by OTP fusewords `SEC_FW_VER0..SEC_FW_VER3`. The OTP fusemap also contains monotonic counter `DCFG_CC_VU` and `IMAGE_KEY_REVOKE` revocation field that can be updated as these are implemented as redundant 16-bit fusewords. See more in i.MX700_Fusemap.xlsx attached in [\[6\]](#).

The dual image boot and secure boot from the FlexSPI interface provides recovery capability for the device. See more in [Section 3.2.1.4](#) and [Section 3.2.5.6](#).

Furthermore, a trust provisioned private key, together with other pre-installed key material, is then used for authentication and secure connection to the device, enabling secure provisioning of OEM assets in manufacturing environment the OEM may not fully trusts. See also [Section 3.2.1.3](#).

113

3.8.3.3 Work unit verdict and rationale

114

The evaluator concludes that the SFRs "Verification of Platform Identity" and "Secure Update of Platform" are included.

115

Therefore, this work unit is fulfilled.

3.8.3.4 Vulnerability analysis notes

116

None.

3.8.4 Work unit ASE_REQ.3-4

ASE_REQ.3.4C: The SARs shall be an exact SESIP assurance level. No augmentation is allowed.

Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	31-03-2026
Verdict:	PASS

3.8.4.1 Related Observation Reports and their status

117 None.

3.8.4.2 Work unit summary of findings

118 In Chapter 4 of [GP_FST_070] defines the package of assurance requirements required for SESIP assurance level 3.

Table 4-3: SESIP3 Assurance Requirements

Assurance Class	Assurance Families
ASE: Security Target evaluation	<i>ASE_INT.1 ST Introduction</i> <i>ASE_OBJ.1 Security requirements for the operational environment</i> <i>ASE_REQ.3 Listed security requirements</i> <i>ASE_TSS.1 TOE summary specification</i>
ADV: Development	<i>ADV_FSP.4 Complete functional specification</i> <i>ADV_IMP.3 Complete mapping of the implementation representation of the TSF to the SFRs</i>
AGD: Guidance documents	<i>AGD_OPE.1 Operational user guidance</i> <i>AGD_PRE.1 Preparative procedures</i>
ALC: Life-cycle support	<i>ALC_CMC.1 Labelling of the TOE</i> <i>ALC_CMS.1 TOE CM coverage</i> <i>ALC_FLR.2 Flaw reporting procedures</i>
ATE: Tests	<i>ATE_IND.1 Independent testing: conformance</i>
AVA: Vulnerability Assessment	<i>AVA_VAN.3 Focused vulnerability analysis</i>

119

120 The mapping and sufficiency rationales declared in the [ST_06] are:

4 Mapping and Sufficiency Rationales

4.1 SESIP3 Sufficiency for the SoC

Table 11. SESIP3 Sufficiency for the SoC

Assurance Class	Assurance Family	Covered By	Rationale
ASE: Security target evaluation	ASE_INT.1 ST Introduction	Section 1	The ST reference is in Section 1.1 , the TOE reference in Section 1.3 , the TOE overview and description in Section 1.6 .
	ASE_OBJ.1 Security requirements for the operational environment	Section 2	The objectives for the operational environment in Section 2 refer to the guidance documents.
	ASE_REQ.3 Listed security requirements	Section 3	All SFRs in this ST are taken from [1]. SFR "Verification of Platform Identity" is included. SFR "Secure Update of Platform" is included. The SARs are an exact SESIP assurance level. No multiple assurance level is claimed.
	ASE_TSS.1 TOE Summary Specification	Section 3	All SFRs are listed per definition, and for each SFR the implementation and verification is defined in the SFR.
ADV: Development	ADV_FSP.4 Complete functional specifications	Material provided to evaluator.	The evaluator will determine whether the provided evidence is suitable to meet the requirement.
	ADV_IMP.3 Complete mapping of the implementation representation of the TSF to the SFRs	Material provided to evaluator.	The evaluator will determine whether the provided evidence is suitable to meet the requirement.
AGD: Guidance documents	AGD_OPE.1 Operational user guidance	Section 1.4	The evaluator will determine whether the provided evidence is suitable to meet the requirement.
	AGD_PRE.1 Preparative procedures	Section 1.4	The evaluator will determine whether the provided evidence is suitable to meet the requirement.
ALC: Life-cycle support	ALC_CMC.1 Labelling of the TOE	Material provided to evaluator.	The evaluator will determine whether the provided evidence is suitable to meet the requirement.
	ALC_CMS.1 TOE CM Coverage	Material provided to evaluator.	The evaluator will determine whether the provided evidence is suitable to meet the requirement.
	ALC_FLR.2 Flaw reporting procedures	Section 3.1.1	The flaw reporting and remediation procedure is described.
ATE: Test	ATE_IND.1 Independent testing: conformance	Material provided to evaluator.	The evaluator will determine whether the provided evidence is suitable to meet the requirement.
AVA: Vulnerability assessment	AVA_VAN.3 Focused vulnerability analysis	N.A. A vulnerability analysis is performed by the	The evaluator performs penetration testing, to confirm that the potential vulnerabilities cannot be exploited in

121

Table 11. SESIP3 Sufficiency for the SoC...continued

Assurance Class	Assurance Family	Covered By	Rationale
		evaluator to ascertain the presence of potential vulnerabilities.	the operational environment for the TOE. Penetration testing is performed by the evaluator assuming an attack potential of Enhanced-Basic.

122

3.8.4.3 Work unit verdict and rationale

123 The evaluator concludes that the SARs are an exact SESIP assurance level.

124 Therefore, this work unit is fulfilled.

3.8.4.4 Vulnerability analysis notes

125 None.

3.8.5 Work unit ASE_REQ.3-5

ASE_REQ.3.5C: If multiple SESIP assurance levels are claimed, it shall be clear to the reader of the ST what SFRs are covered by what SESIP assurance level.	
Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.8.5.1 Related Observation Reports and their status

126 None.

3.8.5.2 Work unit summary of findings

127 None.

3.8.5.3 Work unit verdict and rationale

128 The evaluator concludes that this work unit is not applicable and, therefore, is considered satisfied.

3.8.5.4 Vulnerability analysis notes

129 None.

3.9 Evaluator action ASE_TSS.1.1E

Verdict	PASS
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130 The evaluator shall confirm that the information provided meets all requirements for content and presentation of evidence.

3.9.1 Work unit ASE_TSS.1-1

ASE_TSS.1.1C: The TOE summary specification shall describe how the TOE meets each SFR.
The evaluator <i>shall examine</i> the TOE summary specification to determine that it describes how the TOE meets each SFR.

Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026
Verdict:	PASS

3.9.1.1 Related Observation Reports and their status

131 None.

3.9.1.2 Work unit summary of findings

132 TOE summary specification (TSS) section is not provided in the [ST_06], but as each SFR has a conformance rationale it is satisfied.

3.9.1.3 Work unit verdict and rationale

133 The evaluator concludes that it describes how the TOE meets each SFR.

134 Therefore, this work unit is fulfilled.

3.9.1.4 Vulnerability analysis notes

135 None.

3.10 Evaluator action ASE_TSS.1.2E

Verdict	PASS
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136 The evaluator shall confirm that the TOE summary specification is consistent with the TOE overview and the TOE description.

3.10.1 Work unit ASE_TSS.1-2

The evaluator <i>shall examine</i> the TOE summary specification to determine that it is consistent with the TOE overview and the TOE description.	
Evaluator(s):	JMHO
Trainee evaluator(s):	-
Date:	26-03-2026

Verdict:	PASS
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3.10.1.1 Related Observation Reports and their status

137 None.

3.10.1.2 Work unit summary of findings

138 None.

3.10.1.3 Work unit verdict and rationale

139 The evaluator concludes that the platform overview, platform summary and conformance rationale of each SFR describe the TOE and are consistent.

3.10.1.4 Vulnerability analysis notes

140 None.

4 Observation Reports

141 No observation report has been made during this activity.

5 Vulnerability Analysis

142 No annotation for the vulnerability analysis has been made during this activity.

6 Evidence List

143 The evidences are listed in the document [NXP_iMXRT700-EL_v1.0].